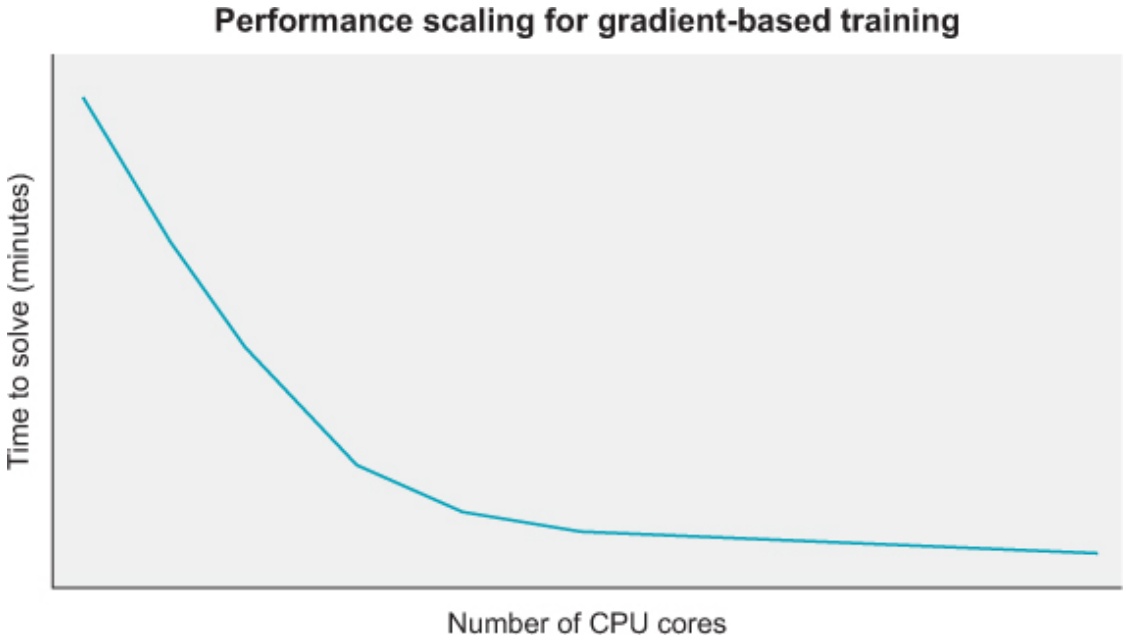
6.5.6. Scaling gradient-based approaches

Gradient-based approaches can be trained on multiple machines as well. However, they do not scale nearly as well as ES. Currently, most distributed training of gradient-based approaches involves training the agent on each worker and then passing the gradients back to a central machine to be aggregated. All the gradients must be passed for each epoch or update cycle, which requires a lot of network bandwidth and strain on the central machine. Eventually the network gets saturated, and adding more workers does not improve training speed as well (figure 6.17).

Figure 6.17. The performance of current gradient-based approaches looks like this. In the beginning, there is a seemingly linear trend because the network has not been saturated. But eventually, as more resources are added, we get less and less of a performance boost.



Evolutionary approaches, on the other hand, do not require backpropagation, so they do not need to send gradient updates to a central server. And with smart techniques like the ones that OpenAI developed, they may only need to send a single number.

Summary